

SkillTwin AI

Multi-Step Prompt Workflow for Predictive Skill-Gap Analysis

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Assessment Type: Mini Assessment – Prompt Workflow Design

1. Project Objective

The idea behind SkillTwin AI is to build a multi-step prompt workflow that creates a kind of digital "skill twin" for a student. This twin is basically a structured picture of the skills a student already has, built from their resume, projects and interests. Once that profile is ready, the workflow goes a step further and tries to predict what skills the student will actually need in the near future, instead of just comparing them to one job description. The whole thing works as a chain of prompts, where each step takes the output of the previous one and builds on it.

2. Problem Statement

Most students know what skills they currently have, but they don't really know what skills will matter two or three years down the line. Job markets and technology keep changing quickly, and a normal resume or a single skill check doesn't really capture that. If you ask a general AI prompt for career advice, you usually get something generic. SkillTwin AI tries to fix this by breaking the problem into smaller, connected steps — first understanding the student, then understanding where the industry is heading, and only then figuring out what to learn.

3. Workflow Architecture

Input Profile → Skill Twin Construction → Future Role & Trend Analysis → Skill Demand Forecasting → Gap Projection → Adaptive Learning Roadmap → Skill Twin Update Plan

4. Workflow Steps and Expected Outputs

Here's a quick summary of what each step does and what comes out of it:

Step	Purpose	Expected Output
1. Skill Twin Construction	Build a structured digital profile of the student using their resume, projects, certifications and stated interests.	Skill twin profile with current skills, proficiency levels and evidence.
2. Future Role & Trend Analysis	Look at where the student's domain is heading — emerging roles, tools and technologies.	List of emerging roles and trending skill clusters.
3. Skill Demand Forecasting	Predict which specific skills are likely to be in high demand over the next 1–3 years.	Ranked future-skill forecast with confidence levels.
4. Gap Projection	Compare the skill twin with the forecasted skills to see what's missing, now and going forward.	Future skill-gap matrix (current vs. projected demand).
5. Adaptive Learning Roadmap	Turn the gaps into an actual step-by-step learning plan.	Roadmap with phases, milestones and project ideas.
6. Skill Twin Update Plan	Decide how often the twin should be refreshed as things change.	Simple update plan with review checkpoints.

5. Prompt-by-Prompt Design

01. Skill Twin Construction

Prompt: "Analyze the provided resume, project list and stated interests. Extract education, experience, technical skills, tools, certifications and demonstrated strengths. Assign an approximate proficiency level to each skill based on the evidence given. Distinguish explicitly stated skills from assumptions. Do not consider future roles yet."

Expected Output: A structured skill twin profile with current skills, proficiency levels and evidence.

02. Future Role and Trend Analysis

Prompt: "Given the student's domain and career direction, identify emerging roles, technologies and industry shifts relevant to that domain over the next 1–3 years. Group findings into trending skill clusters. Do not compare with the student's profile yet."

Expected Output: A list of emerging roles and trending skill clusters for the domain.

03. Skill Demand Forecasting

Prompt: "Using the trend analysis, rank which specific skills are most likely to be in high demand for the student's target roles. Assign a confidence level to each prediction and briefly justify it."

Expected Output: A ranked future-skill forecast with confidence levels and reasoning.

04. Gap Projection

Prompt: "Compare the skill twin profile with the skill demand forecast. Classify each forecasted skill as Matched, Partially Matched or Missing. Cite the relevant evidence for each classification. Do not invent experience."

Expected Output: A future skill-gap matrix comparing current capability with projected demand.

05. Adaptive Learning Roadmap

Prompt: "Using the gap projection, build an ordered, phased learning roadmap. For each phase include a learning objective, a practice task, a small project idea and a measurable completion criterion."

Expected Output: A phased roadmap with milestones, practice tasks and project ideas.

06. Skill Twin Update Plan

Prompt: "Using all previous outputs, define how often the skill twin should be refreshed and what should trigger an update — a new project, a certification, or a shift in industry trends."

Expected Output: A simple, repeatable update protocol with review checkpoints.

6. Prompt Chaining and Information Flow

Each step in this workflow feeds into the next one. Step 1 builds the skill twin. Step 2 looks at the future of the domain on its own, without being influenced by the student's current skills. Step 3 turns that trend analysis into an actual forecast of which skills matter most. Step 4 compares this forecast against the skill twin to find the real gaps. Step 5 converts those gaps into a learning roadmap, and Step 6 makes sure the whole thing doesn't go stale. This chaining is what makes the workflow feel more like a process than a single one-off answer.

Stage	What Gets Passed Forward	What It Enables
1 → 2	Skill twin profile	Study future roles without bias from current skills.
2 → 3	Emerging roles + trend clusters	Forecast which specific skills matter most.
3 → 4	Skill demand forecast + skill twin	Detect which forecasted skills are missing or weak.
4 → 5	Future skill-gap matrix	Build a phased, priority-ordered learning path.
5 → 6	Roadmap + all previous findings	Design a repeatable protocol to keep the twin current.

7. Demonstration Example

Input Profile: A pre-final-year Computer Science student aiming for a career in Data and AI, currently knows Python, SQL and has done one basic machine learning project using scikit-learn. No cloud experience yet.

Stage	Example Result
Skill Twin Construction	Current skills: Python, SQL, basic ML (scikit-learn), one data-analytics project, no cloud experience.
Future Role & Trend Analysis	Emerging roles: ML Engineer, Data Platform Engineer, AI Product Analyst. Trend clusters: MLOps, cloud-native pipelines, GenAI tooling.
Skill Demand Forecasting	High-confidence rising skills: cloud deployment (AWS/GCP), MLOps basics, vector databases, prompt engineering.
Gap Projection	Matched: Python, SQL. Partial: ML fundamentals → MLOps. Missing: cloud deployment, vector databases, prompt engineering.
Adaptive Learning Roadmap	Phase 1: cloud fundamentals. Phase 2: MLOps + model deployment project. Phase 3: vector database + GenAI mini-project.
Skill Twin Update Plan	Refresh the twin every 8–10 weeks or after finishing a project/certification; re-run forecasting once a semester.

8. Expected Learning Outcome

This workflow shows how a career-planning task can be broken down into smaller, connected AI prompts instead of relying on one big generic prompt. It touches on profile building, trend analysis, forecasting, gap comparison, roadmap planning and keeping things updated over time. The focus throughout is on staying evidence-based rather than making things up.

9. Conclusion

Overall, SkillTwin AI takes a student's current profile and turns it into something more useful — a living "skill twin" that's compared against where the job market is actually heading, not just against a single job post. By splitting the work into six connected steps — building the twin, studying trends, forecasting demand, projecting gaps, building a roadmap and updating the twin — the workflow gives a more structured and forward-looking way to plan a career, one that can keep adapting as both the student and the market change.